

Streaming and Batch Audit of Data Quality

1. Overview

Even if systems are online and APIs return **200 OK**, data can still be wrong.

Examples:

- Missing rows after ETL
- Duplicate orders
- Incorrect balances
- Broken foreign keys
- Delayed event processing
- Partial writes

This is a **data quality problem**.

So modern systems need **data audits**.

Availability tells you service is up.
Data quality tells you system is trustworthy.

2. What is Data Quality?

Data quality means data is:

- Accurate
- Complete
- Consistent
- Timely
- Valid
- Reliable for business use

3. Example of Poor Data Quality

None

1000 payments succeeded

Only 992 rows inserted into reporting DB

Missing 8 rows = silent corruption.

4. Why Auditing is Needed

Normal request validations may miss issues caused by:

- Async pipelines
- ETL bugs
- Retry duplication
- Schema changes
- Race conditions
- Human mistakes
- Third-party integration failures

5. Two Types of Data Auditing

A. Streaming Audit (Real-Time / Near Real-Time)

Continuously validates fresh data as events flow.

Used when fast detection is required.

Example

None

Payment Success Event



Audit Consumer checks:

- Payment row exists?
- Amount valid?
- Duplicate?

Use Cases

- Payments
- Orders
- Fraud systems
- Inventory updates
- Real-time analytics

B. Batch Audit (Periodic)

Runs on schedule:

- Hourly
- Daily
- Weekly

Scans larger datasets.

Example

None

Every night:

Orders count in OLTP DB

vs

Orders count in Warehouse

Use Cases

- Reporting validation
- Historical consistency
- Reconciliation
- Compliance checks

6. Streaming vs Batch Comparison

Factor	Streaming Audit	Batch Audit
Speed	Seconds/Minutes	Hours/Days
Scope	Recent records	Large history
Cost	Higher	Lower
Use Case	Critical systems	Deep reconciliation

7. Common Audit Checks

Completeness

Expected rows present?

None

100 events produced

100 rows consumed?

Accuracy

Values correct?

None

Order total = sum(items)?

Consistency

Same data across systems?

None

Payment amount in DB = ledger amount

Freshness

Data not stale?

None

Dashboard updated within last 5 min?

Uniqueness

No duplicates?

None

No duplicate transaction_id

Referential Integrity

Child rows match parent rows.

8. Silent Errors

Most dangerous failures:

System looks healthy but data wrong.

Examples:

- 200 OK returned
- Kafka consumer skipped messages
- Dashboard shows stale numbers
- Payment captured twice

Audits detect what monitoring may miss.

9. High-Level Audit Architecture

None

Source DB / Events



Validation Jobs

├— Streaming Checks

└— Batch Checks



Audit Results Store



Dashboards + Alerts

10. Alerting on Audit Failures

Examples:

- Missing rows > 0.1%
- Duplicate IDs detected
- Freshness delay > 10 min
- Balance mismatch > ₹1000

11. Example: E-commerce Audit

None

Orders Created = 10,000

Payments Captured = 9,997

Shipments Created = 9,950

Need investigation.

12. Example: Banking Audit

None

Ledger credits - debits must net to zero

Any mismatch = severe incident.

13. Batch ETL + Audit Together

Pipeline:

None

OLTP DB



ETL Job



Warehouse



Post-load Audit Job

Checks:

- Row count match
- Null % change
- Schema drift
- Value ranges

14. Governance Benefits

Audits also support:

- Compliance
- Financial controls
- SOX / regulations
- Data trustworthiness

15. Common Causes of Data Quality Problems

- Schema mismatch
- Timezone bugs
- Duplicate retries
- Partial transaction commits
- Backfill mistakes
- Late-arriving events
- Human manual updates

16. Common Tools

- Apache Spark
- Kafka Streams

- Flink
- Airflow
- dbt tests
- Great Expectations
- SQL validation jobs

17. Interview Script

Say:

“I’d add streaming audits for recent critical transactions and nightly batch reconciliation for historical consistency. This helps detect silent failures that normal API monitoring may miss.”

Excellent answer.

18. Final Summary

- Data quality = trustworthy usable data
- APIs can succeed while data is wrong
- Streaming audits detect fast issues
- Batch audits detect deep historical issues
- Alerts should trigger on mismatches
- Essential for payments, finance, analytics

19. Memorable Line

A healthy API with corrupted data is still an outage.